Justin Finkel

Postdoctoral Associate

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Employment

Massachusetts Institute of Technology, Postdoctoral Associate in the department of Earth, Atmospheric, and Planetary Sciences, beginning September 2022.
Advisor: Paul O'Gorman

Education

University of Chicago, Ph.D. in Computational and Applied Mathematics, August 2022.
Thesis topic: Atmospheric extremes through the lens of transition path theory
Advisor: Jonathan Weare (NYU)
Co-advisors: Mary Silber (UChicago), Dorian Abbot (UChicago), Edwin Gerber (NYU)

Washington University in Saint Louis, B.A. in Mathematics and Physics, *Magna Cum Laude*, May 2017.
Thesis Project: Changing World Temperature Statistics
Thesis Advisor: Jonathan Katz

Publications

In Preparation

1. **Justin Finkel** and Paul A. O'Gorman. Rare event sampling on moving targets: fast, transient midlatitude rainstorms and heat waves in a general circulation model. 2025. To be submitted to *Geophysical Research Letters*

2. **Justin Finkel** and Paul A. O'Gorman. Boosting to estimate statistics of tails at conditionally optimal advance split times. 2025. To be submitted to *Nonlinear Processes in Geophysics*

3. William J. M. Seviour, **Justin Finkel**, Philip Rupp, and SNAPSI WG2 Authors. Forecast-based attribution of the role of stratospheric variability in temperature and snow extremes. 2025. To be submitted to *Weather and Climate Dynamics*

4. **Justin Finkel**, William J. M. Seviour, and SNAPSI WG2 Authors. Spatially varying surface responses to large-scale stratospheric nudging: localized forecast-based attribution via extreme value theory. 2025. To be submitted to *Weather and Climate Dynamics*

Published

1. Huan Zhang, **Justin Finkel**, Dorian S. Abbot, Edwin P. Gerber, and Jonathan Weare. Using explainable ai and transfer learning to understand and predict the maintenance of atlantic blocking with limited observational data. *Journal of Geophysical Research: Machine Learning and Computation*, 1(4):e2024JH000243, 2024. e2024JH000243 2024JH000243

2. **Justin Finkel** and Paul A. O'Gorman. Bringing statistics to storylines: Rare event sampling for sudden, transient extreme events. *Journal of Advances in Modeling Earth Systems*, 16(6):e2024MS004264, 2024. Software publicly available in Zenodo repository, in https://doi.org/10.5281/zenodo.10608188

3. **Justin Finkel**, Edwin P. Gerber, Dorian S. Abbot, and Jonathan Weare. Revealing the statistics of extreme events hidden in short weather forecast data. *AGU Advances*, 4(2):e2023AV000881, 2023. Software publicly available in Zenodo repository, https://doi.org/10.5281/zenodo.7675972

4. John Strahan, **Justin Finkel**, Aaron R. Dinner, and Jonathan Weare. Predicting rare events using neural networks and short-trajectory data. *Journal of Computational Physics*, 488:112152, 2023

5. **Justin Finkel**, Robert J. Webber, Edwin P. Gerber, Dorian S. Abbot, and Jonathan Weare. Data-driven transition path analysis yields a statistical understanding of sudden stratospheric warming events in an idealized model. *Journal of the Atmospheric Sciences*, 2022. Software publicly available in Github repository, https://github.com/justinfocus12/SHORT

6. **Justin Finkel**, Robert J. Webber, Edwin P. Gerber, Dorian S. Abbot, and Jonathan Weare. Learning forecasts of rare stratospheric transitions from short simulations. *Monthly Weather Review*, 149(11):3647 – 3669, 2021. Software publicly available in Github repository, https://github.com/justinfocus12/SHORT"

7. **Justin Finkel**, Dorian S. Abbot, and Jonathan Weare. Path properties of atmospheric transitions: Illustration with a low-order sudden stratospheric warming model. *Journal of the Atmospheric Sciences*, 77(7):2327 – 2347, 2020. https://doi.org/10.1175/JAS-D-19-0278.1

8. Predrag Popović, **Justin Finkel**, Mary C. Silber, and Dorian S. Abbot. Snow topography on undeformed arctic sea ice captured by an idealized "snow dune" model. *Journal of Geophysical Research: Oceans*, 125(9):e2019JC016034, 2020. https://doi.org/10.1029/2019JC016034

9. **J. M. Finkel** and J. I. Katz. Changing world extreme temperature statistics. *International Journal of Climatology*, 38(5):2613–2617, 2018. https://doi.org/10.1002/joc.5342

10. **J. M. Finkel** and J. I. Katz. Changing us extreme temperature statistics. *International Journal of Climatology*, 37(13):4749–4755, 2017. https://doi.org/10.1002/joc.5115

11. C.D. Kreisch, J.A. O'Sullivan, R.E. Arvidson, D.V. Politte, L. He, N.T. Stein, **J. Finkel**, E.A. Guinness, M.J. Wolff, and M.G.A. Lapôtre. Regularization of mars reconnaissance orbiter crism along-track oversampled hyperspectral imaging observations of mars. *Icarus*, 282:136–151, 2017. https://doi.org/10.1016/j.icarus.2016.09.033

12. **J. M. Finkel**, L. M. Canel-Katz, and J. I. Katz. Decreasing us aridity in a warming climate. *International Journal of Climatology*, 36(3):1560–1564, 2016. https://doi.org/10.1002/joc.4421

Presentations

1. Yale Atmospheric, Ocean, and Climate Dynamics seminar, October 2024. Invited talk.

2. Cross-VESRI (Schmidt Sciences) convening, July 2024, Cambridge, UK. Contributed talk.

3. Atmospheric and Oceanic Fluid Dynamics meeting, June 2024. Poster.
4. American Physical Society March Meeting, 2022. Contributed talk. March 14, 2022.
5. Center for Atmosphere Ocean Science, New York University Courant Institute seminar. Joint with Jonathan Weare. October 13, 2021
6. University of Bristol Climate Dynamics seminar. Joint with Dorian Abbot. May 26, 2021
7. European Geophysical Union General Assembly, 2021. Contributed talk. April 28, 2021
8. SIAM conference on dynamical systems, 2021. Contributed talk. May 23, 2021
9. American Physical Society March Meeting, 2021. Contributed talk. March 18, 2021
10. Courant Institute of Mathematical Sciences student seminar, 2021. March 12, 2021
11. SIAM conference on dynamical systems, 2019. Poster. May 22, 2019
12. American Geophysical Union Fall Meeting, 2018. Contributed talk. December 10, 2018
13. Midstates Consortium for Math and Science, Nov. 5, 2016. Poster. November 5, 2016
14. Washington University Undergraduate Research Symposium. October 2014. Poster.

Teaching and mentorship

1. Supervising undergraduate research project by Jesús Lopez (Texas A&M, San Antonio), summer/fall 2024. Characterizing spatial statistics of extreme events in passive scalar flows and their response to perturbations.
2. Consulting on an undergraduate thesis project by Sarah Packman (Harvard) on sampling extreme El Niño events with a rare event algorithm I developed.
3. Taught a short virtual linear algebra course to 10 beginning chemistry Ph.D. students (UChicago), September 2020.
4. Supervised undergraduate research project by James Butler (UChicago), 2020-2021. Exploring stochastic stability and transition path statistics of a low-order atmospheric blocking model.
5. Supervised a master's thesis by Matthew Shin (UChicago), 2018-2019. "Towards time-dependent transition path theory: numerical study of periodically forced dynamics."
6. Online tutoring in mathematics, physics, and computer science with Varsity Tutors, 2017-2021.

Outreach activities

1. Classroom visits in Boston area and Cambridge Science Festival demonstration with MIT's "Let's Invest in K-12" (LINK-12) program. Computer simulation and fluid dynamics demonstrations in rotating tanks.

Awards

- Department of Energy Computational Sciences Graduate Fellowship (DOE CSGF), 2018-2022
- Nishi Luthra award for "Outstanding students in the Physics department and the Philosophy department", 2017
- *Sigma Pi Sigma* inductee (physics honor society), 2017
- Academic Mentor of the Year (for physics mentoring), 2014-2015

Last updated: July 4, 2025